Electromagnetically Induced Transparency in NV Centers at Elevated Temperature

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We investigate electromagnetically induced transparency in nitrogen-vacancy centers in diamond under elevated-temperature conditions. By modeling the temperature-dependent excited-state dephasing and the breakdown of the effective three-level Λ system, we clarify the parameter regime in which EIT can be distinguished from Autler–Townes splitting. Our results suggest that phonon-induced processes impose a practical limitation on high-temperature EIT in solid-state spin ensembles.

Electromagnetically induced transparency (EIT) is a quantum interference effect that suppresses optical absorption in a three-level Λ system. In nitrogen-vacancy (NV) centers in diamond, EIT provides an all-optical probe of ground-state spin coherence and has potential applications in quantum sensing and optical quantum memories.

Critical to EIT-based application is the ability to maintain the required coherence properitess at room temperatuer. This has motivated the use of atomic gases such as Ge. However, the solid-state is desireble for practical application. EIT in solid-state has been observerd in varioues systems. However, the transmission of many rare-earth doped crystals is week.

Nitrogen-Vancacy(NV) centers in diamond are promising candibates, owing to long spin coherens time in ground states. EIT in diamond has been observerbed, but the EIT condition has been limited at cryogenc temperature.

In this Letter, we report EIT in NV centers can be induced at room temperature. We developed a theoretical model that incorporates temperature effect. By increasing the energy difference between the excited state and the other one we can suppress the breakedown of the effective three-level Λ system.

We study the EIT in NV centers by considering decoherence, spontanoues emission, dephasings, and phonon-induced transition. EIT is observed when the control field is applied and probe field is off-resonant. The state of the system with decoherence can be described by the density matrix ρ that satisfies the master equation.

In 6-levels, the ground states $|g_1\rangle$ and $|g_2\rangle$ respectively, and the excited state $|e\rangle$,$|e\rangle$,$|e\rangle$,$|e\rangle$. The probe field couples $|g_1\rangle$ to $|e\rangle$, while the control field couples $|g_2\rangle$ to $|e\rangle$.

Solving the master equation corresponding to our model, we can obtain the steady-state solution for the density matrix elements. The susceptibility of the medium can be calculated from the coherence term ρ_{eg_1}, which determines the absorption and dispersion properties of the probe field.

When applying the probe and control field, the hamiltonian to describe the interraction can be witten as

$$\begin{aligned}\frac{H}{\hbar} =& -\delta\,|2\rangle\langle 2| - \Delta_p\,|3\rangle\langle 3| - (\Delta_p - \Delta_e)\,|4\rangle\langle 4| \\ &+ \frac{1}{2}\left[\Omega_{p1}\,|3\rangle\langle 1| + \Omega_{p2}\,|4\rangle\langle 1| + \Omega_{c1}\,|3\rangle\langle 2| + \Omega_{c2}\,|4\rangle\langle 2| + \text{H.c.}\right]\end{aligned} \tag{1}$$

where Δ_p and Δ_c are the probe and control detunings, respectively, and$\delta = \Delta_p - \Delta_c$

When increasing the temperature, the phonon-induced transition between the -orbital states becomes significant. This rate can be given by

$$k_{\mathrm{ph}}(T) = \alpha_{\mathrm{ph}} I(\omega_c, T, \delta_\perp) T^5, \tag{2}$$

where α_{ph} is a proportionality constant and $I(\omega_c, T, \delta_\perp)$ is a dimensionless integral factor. The transverse strain splitting is written as $\Delta_\perp = 2d_\perp\sqrt{\delta_x^2 + \delta_y^2}$.

For an ideal three-level Λ system, EIT appears around the two-photon resonance condition

$$\Delta_p = \Delta_c. \tag{3}$$

Master equation is given by

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \Sigma_i L_i \rho L_i^\dagger \tag{4}$$

The approximate condition for observing a transparency window is

$$|\Omega_c| \gtrsim \sqrt{\gamma_{21}\gamma_{31}}, \tag{5}$$

where γ_{21} is the ground-state decoherence rate and γ_{31} is the optical coherence decay rate.

The condition for forming Λ system is caused by Excited-Level Anti Crosing(ESLAC), which occurs when the energy difference between the two excited states is smaller.This energy difference can be controled by appling magnetic field along z-axis, and strain field perpendicular to the z-axis.

The table shows the comparison of EIT conditions with Alkali-atom and NV centersa at room temperature. The

EIT condition is satisfied at room temperature in Ideal Λ system, but in NV center, nearby excited states cause the breakdown of the effective Λ systemm.

The figure shows the calculated EIT spectrum for different temperatures. At low temperature, a clear EIT window is observed. As the temperature increases, more large Ω_c is required.

In 6-level system, the EIT window is observed at low temperature, but as the temperature increases, the EIT window becomes less pronounced and eventually disappears weven when the control field is strong enough to satisfy the EIT contrast.